

Lab 2 – TrashTag Product Specification

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April 3rd, 2026

Version 1

## Table of Contents

|     |                                               |                                     |
|-----|-----------------------------------------------|-------------------------------------|
| 1   | Introduction.....                             | 3                                   |
| 1.1 | Purpose.....                                  | 3                                   |
| 1.2 | Scope.....                                    | 3                                   |
| 1.3 | Definitions, Acronyms, and Abbreviations..... | 4                                   |
| 1.4 | References.....                               | <b>Error! Bookmark not defined.</b> |
| 1.5 | Overview.....                                 | 7                                   |
| 2   | Overall Description.....                      | 7                                   |
| 2.1 | Product Perspective.....                      | 7                                   |
| 2.2 | Product Functions .....                       | 8                                   |
| 2.3 | User Characteristics .....                    | 8                                   |
| 2.4 | Constraints .....                             | 9                                   |
| 2.5 | Assumptions and Dependencies .....            | 9                                   |

## **1 Introduction**

### **1.1 Purpose**

The purpose of this Software Requirements Specification is to define the functional and non-functional requirements for the TrashTag system. This document is intended for developers, testers, and stakeholders involved in system implementation.

### **1.2 Scope**

TrashTag is a browser application designed to improve the waste cleanup process by allowing users to photograph and report litter and organize and conduct cleanups. Live reporting will be available for users to get real-time litter reports and users will be alerted to nearby litter. Images of litter will be provided to users as well as verification of whether the litter is still there or not. A ranking system will be used as gamification for the cleanup process. Users will have access to a leaderboard to see where they rank against other users in their area or even worldwide.

TrashTag allows users to take pictures of trash, pin its location on a map, upload it to the website, and organize clean up groups to help dispose of illegally dumped litter. Local governments will spend less money searching for litter when people in the community can tag it for them, and volunteers will have an easier time organizing events when they are all using the same application. When local governments, non-profits, and volunteers are all connected in one browser application, they can more easily work together to clean up litter.

TrashTag aims to help the environment and the public by providing a resource that clearly highlights the location and type of waste as well as providing live updates regarding existing cleanup efforts and geographic littering patterns to help volunteers allocate their time and resources effectively.

### 1.3 Definitions, Acronyms, and Abbreviations

**API:** Application Programming Interface. A software mechanism for accepting and providing data from and to external applications

**Illegal Dumping:** The unauthorized disposal of trash and other hazardous materials onto private and/or public property.

**Metadata:** Descriptive information within data files such as date, location, and timestamp data in photos to be submitted to TrashTag.

**Geotagging:** The addition of GPS coordinates to images that enable precise location identification where trash was reported.

**Live Real-Time Updates:** Live data synchronization displaying trash reports to both users and cleanup organizations instantaneously.

**Gamification:** The incorporation of game-like mechanics such as rankings to motivate user participation and enhance cleanup operations.

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## 1.5 Overview

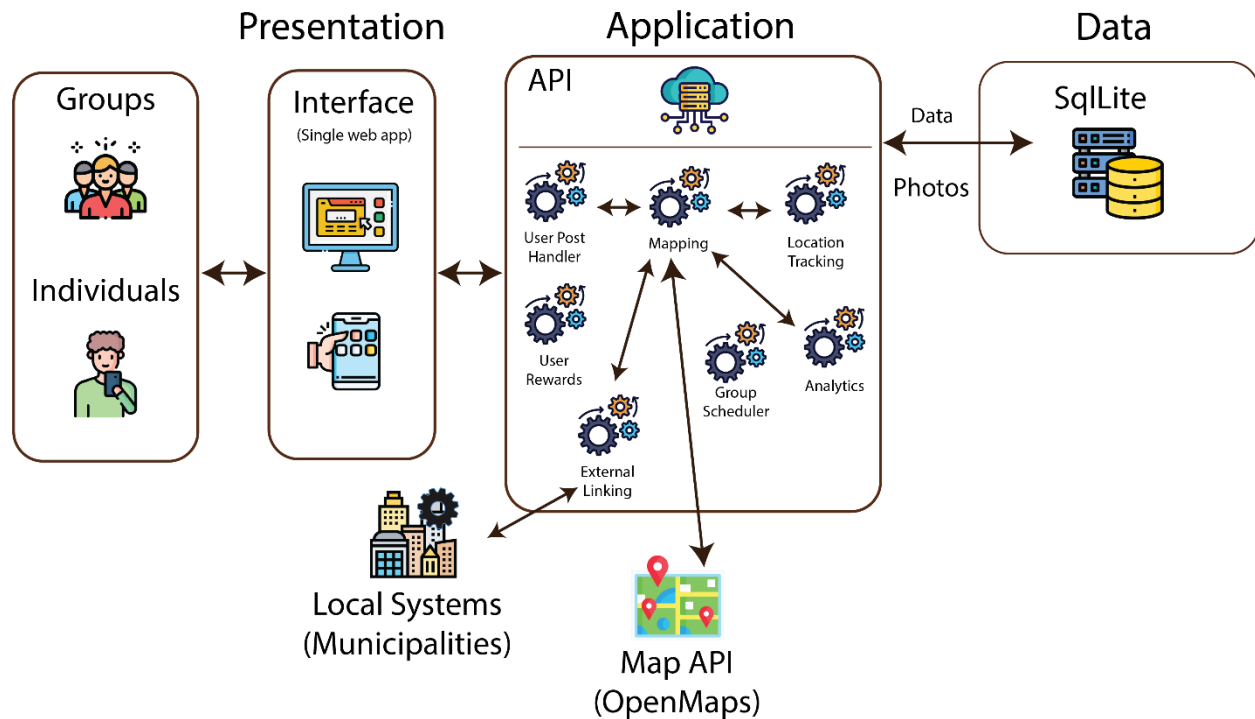
## 2 Overall Description

### 2.1 Product Perspective

TrashTag is a browser application designed to improve the waste cleanup process by allowing users to photograph and report litter and organize and conduct cleanups. Live reporting will be available for users to get real-time litter reports and users will be alerted to nearby litter. Images of litter will be provided to users as well as verification of whether the litter is still there or not. A ranking system will be used as gamification for the cleanup process. Users will have access to a leaderboard to see where they rank against other users in their area or even worldwide.

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to help the environment and the public by providing a resource that clearly highlights the location and type of waste as well as providing live updates regarding existing cleanup efforts and geographic littering patterns to help volunteers allocate their time and resources effectively.



## 2.2 Product Functions

As of now TrashTag has fully implemented Litter Reporting, Mapping, Client Authorization, Cleanup Tracking, and Verification. All other items are being reviewed for future implementation.

## 2.3 User Characteristics

There are four total user roles: Citizens, Organizations, Local Governments, and Moderators.

Citizens are community members who want to help clean up their community by reporting litter-affected areas. They do so by uploading images of litter, marking severity levels, collaborating with other users in cleaning events, and earning rankings as digital rewards for the efforts.



Organizations and Non-Profit Environmental Groups are members that organize cleanup events for users to participate in and allocate proper resources for those events. They also have the ability to embed the app's map widget onto their website, view and download litter trend reports, view density analytics and cleanup statistics, and track cleanup event impact on the community. This role is for government staff to reduce cleanup costs and optimize waste management. Like organizations, they can view litter trends and density heavy areas, as well as gather and download statistics to help with making policy and budget decisions.

For government users, their user information is protected and remains anonymous. Governments would be able to identify pickup areas, allocate resources, advertise cleanup events, and verify events took place.

Finally, Moderators are platform managers who ensure system integrity and quality remain intact. Moderators can view the litter maps and filter them like other users, as well as delegate cleanup tasks and view cleanup events. Unlike other user roles, Moderators can review and respond to flagged reports, edit and correct those reports (if needed), remove duplicate or suspicious reports, and issue violation warnings to users. This ensures that the platform provides accurate data analytics for all its users.

## **2.4 Constraints**

N/A

## **2.5 Assumptions and Dependencies**

N/A